

PENUMBRA

Software Requirements Specification (SRS)

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Product: PENUMBRA — Probabilistic Space-Weather Driver Forecasting

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for **PENUMBRA**, an open-source system that produces **probabilistic forecasts of the space-weather drivers** that govern satellite orbital drag — the F10.7 solar radio flux and the Kp geomagnetic index — attaches calibrated uncertainty to every prediction, verifies that calibration against ground truth, and translates the driver uncertainty into orbital-drag and orbit-decay risk. It is delivered as a REST API and a web dashboard.

1.2 Scope

Atmospheric drag is the largest source of orbit-prediction error in LEO, and the dominant sub-source is frequently the *forecast of the drivers* feeding the density model, not the density model itself (Parker et al., 2024). Operational driver forecasts — including NOAA's 45-day F10.7/Ap product — are published as **point values with no uncertainty quantification**, forcing operators to make unsupported assumptions during orbit and conjunction assessment. Research-grade probabilistic methods exist but there is **no operational product**. PENUMBRA fills that gap. It is the phase-2 companion to the KESSLER conjunction-assessment toolkit; together they form an open orbital-risk stack.

Out of scope for v1.0: full physics-based thermosphere modelling (PENUMBRA uses a documented reduced density model for the impact translation), ionospheric/TEC forecasting, solar-flare prediction, and paid data sources.

1.3 Definitions and Acronyms

Term	Definition
F10.7	Solar radio flux at 10.7 cm (2800 MHz), in solar flux units (sfu). Primary proxy for solar EUV heating of the thermosphere.
Kp	Planetary geomagnetic activity index, quasi-logarithmic scale 0–9 (in thirds), 3-hourly.
Ap	Daily linearised geomagnetic amplitude index derived from Kp.
Driver	An input index (F10.7, Kp/Ap) that drives an atmospheric-density model.
Lead time	Forecast horizon: days ahead of issue date (1...45).
Quantile band	Interval between two forecast quantiles (e.g. p05–p95 = 90% band).
Calibration	Property that a stated X% band contains the truth X% of the time over many forecasts.
Coverage	Empirical fraction of observations falling inside a stated band.
Pinball loss	Proper scoring rule for a quantile forecast (a.k.a. quantile loss).
Skill score	Relative improvement of a forecast over a reference (here, NOAA's 45-day forecast).
SWPC	NOAA Space Weather Prediction Center.

1.4 References

- Parker & Linares (2024) — Influences of Space Weather Forecasting Uncertainty on Satellite Conjunction Assessment, *Space Weather* (AGU), DOI 10.1029/2023SW003818.
- NOAA SWPC data services — `services.swpc.noaa.gov/json/` (F10.7, Kp, 45-day forecast).
- GFZ Potsdam — Kp index archive, `kp.gfz-potsdam.de`.
- Vallado, *Fundamentals of Astrodynamics and Applications* — exponential atmosphere and drag.

2. Overall Description

2.1 Product Perspective

PENUMBRA is a self-contained web application plus a reusable forecasting core. It consumes only free public NOAA/GFZ data and requires no paid dependency. It runs fully on a small VPS; the service layer isolates storage and scheduling behind interfaces so the system can later move to cloud infrastructure without touching the forecasting logic.

2.2 User Classes

User class	Description	Primary interest
Satellite operator	Plans manoeuvres and orbit maintenance.	"How uncertain is my predicted orbit over the next N days, and why?"
Orbit-prediction / SSA software	Consumes driver forecasts programmatically.	A driver <i>distribution</i> , not a point, to propagate state uncertainty honestly.
Space-weather researcher	Studies driver forecasting and calibration.	Reproducible probabilistic forecasts, verification metrics, skill vs NOAA.
Demo viewer	Evaluates the project.	Clear fan charts, the calibration story, polish.

2.3 Operating Environment

- Backend: Python ≥ 3.11 (3.9 supported via typing back-port), FastAPI/uvicorn.
- Frontend: Node ≥ 18 , Vite build, evergreen browsers.
- v1.0 target: localhost and a shared Lightsail VPS (co-hosted with KESSLER, isolated service/port).

2.4 Design Constraints

CON-1	Forecasts MUST carry calibrated uncertainty; a point value without a band is not an acceptable output. Uncertainty is the product.
CON-2	Uncertainty bands MUST be derived from a leakage-free walk-forward backtest (out-of-sample error quantiles), never fit in-sample.
CON-3	The system MUST be fully functional with zero credentials (Demo Mode) using bundled historical data.
CON-4	NOAA/GFZ are public but not to be hammered: cache-first architecture, polite intervals; no per-user live fetches.
CON-5	Repository layout mirrors KESSLER: separate <code>ui/</code> and <code>backend/</code> ; backend separates <code>logic/</code> (pure math) and <code>service/</code> (I/O). Deploy MUST NOT disturb the co-hosted KESSLER service.

3. Functional Requirements

3.1 Data Ingestion (FR-ING)

ID	Requirement	Priority
FR-ING-1	SHALL ingest daily observed F10.7 from NOAA SWPC (<code>f107_cm_flux.json</code>) and long-history monthly indices (<code>solar-cycle/observed-solar-cycle-indices.json</code>), keyless.	Must
FR-ING-2	SHALL ingest planetary Kp from NOAA SWPC (<code>products/noaa-planetary-k-index.json</code>) and derive daily max-Kp / Ap.	Must
FR-ING-3	SHALL ingest NOAA's official 45-day F10.7/Ap forecast (<code>45-day-forecast.json</code>) as the reference baseline for skill scoring.	Must
FR-ING-4	All external fetches SHALL be cached with per-source TTL (observations ≥ 3 h, forecast ≥ 6 h) and served from cache within TTL.	Must
FR-ING-5	In Demo Mode the system SHALL load bundled historical series and mark responses <code>data_mode: "demo"</code> .	Must
FR-ING-6	Ingestion failures SHALL degrade to last-known-good cache with a structured warning, never a hard failure.	Must

3.2 F10.7 Probabilistic Forecast (FR-F107)

ID	Requirement	Priority
FR-F107-1	SHALL produce a point forecast of daily F10.7 for lead times 1–45 days using a persistence-with-mean-reversion model toward the local solar-rotation (27-day) and cycle climatology.	Must
FR-F107-2	SHALL attach quantile bands (at least p05, p25, p50, p75, p95) per lead day, derived from the empirical distribution of out-of-sample forecast errors at that lead time (FR-VER-1).	Must
FR-F107-3	Bands SHALL be monotonic ($p05 \leq p25 \leq \dots \leq p95$) and F10.7 SHALL be floored at the physical quiet-Sun minimum (~ 64 sfu).	Must
FR-F107-4	All forecast math SHALL be pure functions in the logic layer (no I/O) and unit-tested, including monotonicity and a synthetic-signal recovery test.	Must

3.3 Kp / Geomagnetic Forecast (FR-KP)

ID	Requirement	Priority
FR-KP-1	SHALL produce, per lead day 1–7, probabilities across geomagnetic categories: Quiet ($Kp < 3$), Unsettled ($3 \leq Kp < 4$), Active ($4 \leq Kp < 5$), Storm ($Kp \geq 5$).	Must
FR-KP-2	Category probabilities SHALL be estimated from the recency-weighted historical transition frequencies conditioned on recent activity, and SHALL sum to 1 per lead day.	Must
FR-KP-3	SHALL expose the expected daily Ap with an uncertainty band for downstream drag modelling.	Should

3.4 Verification & Calibration (FR-VER)

ID	Requirement	Priority
FR-VER-1	SHALL run a walk-forward backtest over the historical record: at each origin, forecast forward, then record the signed error per lead time — with a strict train/verify time split (no future leakage).	Must
FR-VER-2	SHALL compute band coverage (empirical fraction inside p05–p95 and p25–p75) per lead time; a calibrated 90% band targets 0.90 coverage.	Must
FR-VER-3	SHALL compute a reliability diagram (predicted vs observed frequency) for the Kp categorical forecast.	Must
FR-VER-4	SHALL compute pinball loss for the F10.7 quantiles and a skill score of the PENUMBRA point forecast against NOAA's 45-day forecast (RMSE-based, per lead time).	Must
FR-VER-5	The calibration report SHALL be exposed via API and rendered in the UI — it is the product's primary credibility artifact.	Must

3.5 Drag-Impact Translation (FR-DRAG)

ID	Requirement	Priority
FR-DRAG-1	SHALL map the F10.7 forecast band to a thermospheric density band at reference altitudes (e.g. 400, 550, 800 km) via a documented reduced density model (F10.7-scaled exponential atmosphere).	Must
FR-DRAG-2	SHALL express the resulting along-track position / orbit-decay uncertainty growth for a nominal LEO object over the forecast window — the explicit bridge to conjunction assessment.	Should
FR-DRAG-3	SHALL clearly label the drag translation as reduced-model, order-of-magnitude guidance — not a high-fidelity propagation.	Must

3.6 API (FR-API)

ID	Requirement	Priority
FR-API-1	REST API under <code>/api/v1</code> : F10.7 forecast, Kp forecast, drag impact, calibration report, recent observations, pipeline status, health.	Must
FR-API-2	All errors SHALL use one envelope <code>{"error": {code,message,detail,request_id}}</code> with correct HTTP status.	Must
FR-API-3	OpenAPI schema SHALL be served at <code>/docs</code> .	Must

3.7 Visualisation (FR-UI)

ID	Requirement	Priority
FR-UI-1	The dashboard SHALL render an F10.7 fan chart : observed history, point forecast, and nested p25–p75 / p05–p95 bands over 45 days, with NOAA's forecast overlaid for comparison.	Must
FR-UI-2	SHALL render the Kp storm-probability forecast as a stacked per-day category strip, and the drag-impact panel.	Must
FR-UI-3	A Calibration page SHALL render coverage-by-lead, the reliability diagram, and the skill-vs-NOAA curve.	Must
FR-UI-4	A Pipeline page SHALL embed the n8n-style agent/data-flow diagram with live stage status.	Must
FR-UI-5	Visual design: dark space-grade theme with a solar accent; every chart labelled with units, lead time, and issue date; product description and Apache-2.0 licensing shown in the UI.	Must
FR-UI-6	API failures SHALL surface as non-blocking toasts with retry; charts keep last-good data.	Must

4. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Reliability	Every service entry point wraps external I/O in typed exception handling; no unhandled exception crosses the API boundary (global handler guarantees the error envelope).
NFR-2	Observability	Structured logging with request_id to console and rotating file; each pipeline agent run logged with duration and item counts.
NFR-3	Performance	Cached forecast endpoints p95 < 200 ms; a full walk-forward backtest over the bundled record completes in < 10 s.
NFR-4	Security	No credentials required; any optional keys via environment only; CORS limited to known origins.
NFR-5	Portability / Coexistence	Runs as an isolated systemd service on its own port; MUST NOT modify or contend with the co-hosted KESSLER service; MemoryMax bounded for the shared 414 MB host.
NFR-6	Maintainability	Logic layer framework-free pure Python; ≥ 85% unit coverage on <code>logic/</code> ; UI strict TypeScript.
NFR-7	Compliance	NOAA/GFZ attribution rendered in UI and API root; NOAA data noted as public domain.

5. Acceptance Criteria

- All Must requirements implemented; QA report (PEN-QA-001) shows tests passing.
- Backend + UI run on localhost in Demo Mode with a working F10.7 fan chart, Kp strip, and calibration page — no console errors.
- Unit tests green: forecast monotonicity + synthetic recovery, backtest leakage-free, coverage/pinball/skill math, drag translation, API contract.
- Documents delivered: SRS, HLD, LLD, QA (PDF) + interactive pipeline diagram (HTML) + video demo.
- Deployed to the shared VPS on an isolated port with KESSLER unaffected.